

DEPARTMENT OF ROBOTICS & ARTIFICIAL INTELLIGENCE

Total Marks: 7.5

Obtained Marks: _____

PFAI-LAB

Project

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“Data Cleaning Report Using Pandas”

Introduction

In this project, a raw sales dataset containing customer orders was cleaned using the Python library Pandas. The original dataset contained multiple issues such as missing values, spelling mistakes, duplicate records, invalid dates, negative values, inconsistent formatting, and incorrect totals.

The main objective of this task was to transform the unclean dataset into a clean and structured format suitable for data analysis.

Problems Found in the Raw Dataset

The dataset initially contained many data quality issues, including:

- Duplicate records
- Extra spaces in customer names and categories
- Incorrect gender values
- Spelling mistakes in product categories
- Quantity values written in words
- Negative ages and prices
- Incorrect total amounts
- Invalid and inconsistent date formats
- Missing values in important columns
- Special characters in customer names
- Inconsistent payment methods

These problems made the dataset unreliable for analysis.

Data Cleaning Process Using Pandas

The following steps were performed to clean the dataset:

1. Removing Duplicate Rows

Duplicate rows were removed to avoid repeated records in the dataset.

Before Cleaning

- Same customer records appeared multiple times.

After Cleaning

- Duplicate entries were removed successfully.

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2. Removing Extra Spaces

Unnecessary spaces were removed from text columns such as customer names and product categories.

Before Cleaning

- " Lily Sharma "
- " Electornics "

After Cleaning

- "Lily Sharma"
- "Electronics"

3. Standardizing Gender Values

Different spellings and formats of gender values were standardized.

Before Cleaning

- M
- F
- femlae
- Femle
- MALE
- FEMALE

After Cleaning

- Male
- Female

4. Correcting Product Categories

Incorrect and inconsistent product category names were corrected.

Before Cleaning

- clothng
- Cloth
- elecronics
- Electornics
- FURNITRE
- grocey

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After Cleaning

- Clothing
- Electronics
- Furniture
- Groceries

5. Converting Quantity into Numeric Values

Text-based quantities were converted into numbers.

Before Cleaning

- one
- two

After Cleaning

- 1
- 2

6. Cleaning the Age Column

Text values and negative ages were corrected.

Before Cleaning

- thirty
- forty
- -35
- -14

After Cleaning

- 30
- 40
- 35
- 14

7. Fixing Negative Prices

Negative prices were converted into positive values.

Before Cleaning

- -299
- -2500

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After Cleaning

- 299
- 2500

8. Recalculating Total Amounts

The Total column was recalculated using the formula:

$$\text{Total} = \text{Quantity} \times \text{Price}$$

Before Cleaning

- Incorrect totals
- Missing totals

After Cleaning

- Accurate totals generated automatically.

9. Standardizing Payment Methods

Payment methods with inconsistent spellings were standardized.

Before Cleaning

- cash
- upi
- debit
- Credit Card

After Cleaning

- Cash
- UPI
- Debit Card
- Credit Card

10. Converting Dates into Standard Format

Different date formats were converted into a proper datetime format.

Before Cleaning

- 07-29-2023
- 25-Jul-2023
-

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- 2023/07/02
- invalid

After Cleaning

- Dates converted into a consistent format
- Invalid dates handled properly

11. Removing Special Characters from Names

Special characters were removed from customer names.

Before Cleaning

- Ish@ Fern@ndez
- Thom@s
- R@vi

After Cleaning

- Ish Fernndez
- Thomas
- Ravi

12. Handling Missing Values

Missing values were either filled or removed depending on the importance of the column.

Before Cleaning

- Missing gender values
- Missing payment methods
- Missing ages and quantities

After Cleaning

- Important missing values handled successfully.

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• **UNCLEANED DATA BEFORE USING PANDAS:**

	A	B	C	D	E	F	G	H	I	J	K	L	M
1	Order_ID	Customer	Gender	Age	Product_C	Quantity	Price	Total	Order_Dat	Payment_Mode			
2	1046	Asha Iyer	FEMALE		20 clothing	3	-299	-897	07-29-2023				
3	1079	Lily Thom	femlae		45 Cloth	one	499	499	#####	cash			
4	1054	Anne Smit	Femle		50 Electronic	2	-299	-598	#####	Credit Card			
5	1061	Usa Smith	F		58 FURNITRE	1	12000	12000	#####	Paytm			
6	1061	Lily Sharr	M		69 Electronic	2	12000	24000	#####	Paytm			
7	1046	Jane Verr	MALE		20 electronics	3	199	597	#####	Cash			
8	1013	Usa doe	female		50 Clothing	3	999	2997	#####	Credit Card			
9	1013	robert cha	femlae		53 clothing	4	1500	6000	invalid	Credit Card			
10	1058	Jane Kun	Female		59 Electronic	2	12000	24000	#####	debit			
11	1092	NOAH NAI	MALE		69 Clothing	one	1500	1500	07-29-2023	Netbanking			
12	1023	Asha Iyer	Female		68 home dec	3	2500	7827	#####	Credit Card			
13	1043	Isha Thom	female		26 Electronic two		999		#####	Debit Card			
14	1096	Isha Gup	femlae		52 clothing	one	12000	12000	#####	Online			
15	1033	anne gupt	Female		36 home dec	3	299		#####	cash			
16	1091	Olivia Kha	FEMALE		61 Electronic	4	999	3996	#####	Online			
17	1014	Asha Thc	Female		-35 Electronic	4	1500	6000	07-18-2023	Credit Card			
18	1046	Jane Verr	MALE		20 electronics	3	199	597	#####	Cash			
19	1035	ravi patel			24 Electronic	4	1500	6000	#####	UPI			
20	1087	Ravi Fem	male		-14 home dec	1	999	999	#####	debit			
21	1005	ANNE THC	M		42 home dec	4	-12000	-48000	#####	Cash			
22	1072	DAVID SH	female	thirty	Grocey	2	2500		#####	Credit Card			
23	1073	Ish@ Fern	Male		20 Clothing	2	5000	10000	25-Jul-23	Online			
24	1077	Emm@ Jol	Female		Groceries two		2500		#####				
25	1059	anne shar	M		grocey	4	299	1196	#####	cash			
26	1001	S@r@ Fern	Female		28 electronics	1	499	499	#####				
27	1008	priya pate	Male		19 Grocey	4	199	796	#####	Cash			
28	1094	Anne Fern	F		37 Clothing	one	299		#####	Online			
29	1099	Maya Guj	M		28 Furniture	one	5000	5000	#####	Cash			
30	1083	Isha Iyer	MALE		34 Furnitre	3	299	897	#####	cash			
31	1014	Robert Do	female		50 Cloth	4	0	0	2023.08.05	upi			
32	1051	USA KHAN	Male		26 clothing	1	499	499	#####	Netbanking			
33	1014	Priya Kur	MALE		-25 Cloth	3	-2500	-7500	#####	Cash			
34	1041	Arun Johi	Male		23 Grocey	3	199	597	#####	Cash			
35	1049	Jane Sharma			-33 electronics	4	499	1996	05/08/2023	cash			

	A	B	C	D	E	F	G	H	I	J	K	L	M
35	1049	Jane Sharma			-33 electronics	4	499	1996	05/08/2023	cash			
36	1037	Lily Johnson			31 Furniture	4	12000	48000	2023.08.23	UPI			
37	1021	Asha Pate	Female		32 Furniture	2	999	1998	#####	cash			
38	1002	Ish@ Sing	male	forty	Clothing	1	12000	12000	#####	Credit Card			
39	1070	Emma Guj	F		28 Cloth	3	299	897	#####	Credit Card			
40	1052	Jane Verma			50 Grocey	3	999	2997	#####	Online			
41	1061	DAVID DO	F		44 clothing	4	999	3996	#####	debit			
42	1061	david johr	femlae		grocey	one	199		#####	Netbanking			
43	1006	noah nair	FEMALE		69 Furnitre	one	199		#####	Online			
44	1089	PRIYA DO	female		43 grocey	1	1500	1500	#####	Paytm			
45	1062	Priy@ Veri	femlae		19 Electronic	one	0	0	#####	upi			
46	1020	Arun Singl	M		-32 Groceries	3	1500	4500	#####	Credit Card			
47	1059	Lis@ Iyer	MALE	thirty	HOME DEC	2	499	998	08-21-2023	Credit Card			
48	1077	OLIVIA VEI	M		-18 grocey	4	5000	20000	16-Jul-2023	Cash			
49	1039	ASHA SINC	F		-4 grocey	4	499	1996	#####	Cash			
50	1091	arun john	MALE		41 Electronic	3	-499	-1649	#####	Paytm			
51	1052	Olivia Tho	M		44 Clothing	1	2500		#####	Online			
52	1047	Lily Patel	MALE		65 GROCERIE	2	1500	3000	#####	Online			
53	1050	tom thom	FEMALE		26 FURNITRE	4	12000	48000	#####				
54	1003	Ish@ Johr	male		66 Furnitre	4	999	3996	#####	debit			
55	1050	Usa Pate	MALE		49 clothing	2	1500	3000	#####	Cash			
56	1079	Arun Thor	Female		-56 electronics	3	299	897	#####				
57	1088	Lily Doe			45 electronics	4	999		04/07/2023	Credit Card			
58	1057	karan sharma			40 grocey	2	5000	10000	#####	Credit Card			
59	1079	Lily Thom	(femlae		45 Cloth	one	499	499	#####	cash			
60	1063	olivia kum	male		-44 Clothing	2	12000	24000	#####	upi			
61	1081	Olivia Sing	Male		24 home dec	1	1500	1500	20-Jul-23	cash			
62	1032	robert sha	F		69 grocey	4	2500	10000	invalid	upi			
63	1060	tom patel	Male		25 Grocey	1	299	299	#####	upi			
64	1034	John Sh@i	Female		53 home dec	2	5000	10000	#####	UPI			
65	1053	KARAN FEI	female	forty	Electornics		5000	5000	04-Jul-23	Credit Card			
66	1038	Ravi Gupti	female		20 home dec	1	199	199	#####				
67	1063		MALE		43 Clothing	2	199	398	11/07/2023				
68	1071	Emm@ Sn	MALE		41 Groceries	3	499	1800	#####	Cash			
69	1090	Sara Iyer	male		39 home dec	4	1500	6000	#####	upi			
70	1020	Emma Verma			66 HOME DEC	one	2500		#####	cash			

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	A	B	C	D	E	F	G	H	I	J	K
69	1090	Sara Iyer	male	39	home deco	4	1500	6000	#####	upi	
70	1020	Emma Verma		66	HOME DEC one		2500		#####	cash	
71	1089	Lily Singh		49	Furnitre	4	499	1996	07-22-202	debit	
72	1071			22	electronics	4	199		invalid	Credit Card	
73	1091	brun johns	MALE	41	Electronic	3	-499	-1649	#####	Paytm	
74	1051	ANNE IYER	MALE	62	HomeDeco	2	199	398	13-Aug-20	cash	
75	1084	Ish@Thom	Male	twenty	Cloth	4	299	1196	invalid	cash	
76	1052	Lis@Smith	FEMALE	49	Cloth	2	2500	5000	#####	Credit Card	
77	1059	Priya Iyer	Female		clothing	3	1500	4500	#####	Credit Card	
78	1059	Arun Ch@	male	51	home deco	2	5000	10000	#####		
79	1043	NOAH PAT	Femlae	33	Grocey	4	1500	6000	08-29-202	UPI	
80	1014	TOM KHAN	Male	52	HOMEDEC	1	1500	1500	#####	Cash	
81	1088	Lily Doe		45	electronics	4	999		04/07/202	Credit Card	
82	1071	NOAH VER	male	41	GROCEY	4	199	796	#####	debit	
83	1080	Mike Kuma	female	36	electronics	1	1500	1500	2023.07.2	Credit Card	
84	1071	Robert Tho	M	35	Furnitre	1	499	499	07-22-202	debit	
85	1060	Arun Sh@r	F	61	home deco one		499		2023.08.0	Credit Card	
86	1048	EMMAPAT	MALE	24	electronics	1	299	299	#####		
87	1088	Jane Doe	F	53	Groceries	1	2500	2500	invalid	Online	
88	1092	RAVI JOHN	MALE	51	CLOTH	2	299	598	#####	debit	
89	1061	MIKE GUPT	F	50	Groceries	3	199	597	#####		
90	1029	Arun Singh	female	22	grocey one		199	199	#####	Credit Card	
91	1001	RAVI IYER	Male	26	GROCEY	4	12000	48000	19-Jul-23	Debit Card	
92	1017	Robert Sha	Female	66	Cloth	1	12000	11671	#####	Cash	
93	1075	lisa verma	Female	21	Groceries	3	5000	15000	#####	UPI	
94	1092	Emm@Gu	male	34	Cloth	1	5000	5000	#####	Debit Card	
95	1003	Ravi Verma	Female	56	Clothing	2	2500	5000	08-15-202	upi	
96	1053	Sara Smid	F	thirty	Furniture	4	299	1196	#####	Credit Card	
97	1092	maya checko		-15	Clothing	2	999	1998	#####	Debit Card	
98	1001	Sara Gupta	M	52	grocey	3	1500	4500	07-23-202	cash	
99	1007	janie kumar	Female	23	Electronics	2	499	998	13-Jul-23	Paytm	
100	1017	Lily Verma		29	Cloth	4	5000	20000	#####	Paytm	
101	1051	Noah Vern	MALE		ELECRON	4	299	1196	#####	Debit Card	
102	1003	Jane Iyer	male	52	Furniture two		299		#####	Netbanking	
103	1060	Emma Fernandez	Female	58	HomeDeco one		299		#####	Credit Card	
104	1001	R@vi Fern@	Female	18	home deco	3	999	2997	#####	Cash	
105	1014	Lisa Patel	Male		Groceries	1	5000	5000	#####		
106	1061	DAVID DOE	F	44	clothing	4	999	3996	#####	debit	

• **REMOVING DULICATES:**

```
import pandas as pd
```

```
df = pd.read_csv("unclean_sales_data (1).csv")
print(df)
df = df.drop_duplicates()
```

	Order_ID	Customer_Name	Gender	Age	Product_Category	Quantity	\
0	1046	Asha Iyer	FEMALE	20	clothing	3	
1	1079	Lily Thom@s	femlae	45	Cloth	one	
2	1054	Anne Smith	Femle	50	Electronics	2	
3	1061	Lisa Smith	F	58	FURNITRE	1	
4	1061	Lily Sharma	M	69	Electornics	2	
..	...	...	...	...	...	...	
100	1003	Jane Iyer	male	52	Furniture	two	
101	1060	Emma Fernandez	Female	58	HomeDecor	one	
102	1001	R@vi Fernandez	Femle	18	home decor	3	
103	1014	Lisa Patel	Male	NaN	Groceries	1	
104	1061	DAVID DOE	F	44	clothing	4	

	Price	Total	Order_Date	Payment_Mode
0	-299	-897.0	07-29-2023	NaN
1	499	499.0	2023/07/02	cash
2	-299	-598.0	2023/07/07	Credit Card
3	12000	12000.0	22/07/2023	Paytm
4	12000	24000.0	17/07/2023	Paytm
..	...	...	...	...
100	299	NaN	2023-07-22	Netbanking
101	299	NaN	18-Aug-2023	Credit Card
102	999	2997.0	2023/07/24	Cash
103	5000	5000.0	07-01-2023	NaN
104	999	3996.0	20/08/2023	debit

[105 rows x 10 columns]

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• REMOVING EXTRA SPACES:

```
import pandas as pd

df = pd.read_csv("unclean_sales_data (1).csv")
print(df)
df = df.drop_duplicates()

df = df.apply(lambda x: x.str.strip() if x.dtype == "object" else x)

Order_ID    Customer_Name  Gender  Age  Product_Category  Quantity  \
0          1046      Asha Iyer  FEMALE  20         clothing         3
1          1079      Lily Thom@s  femlae  45          Cloth         one
2          1054      Anne Smith  Femle  50      Electronics         2
3          1061      Lisa Smith    F  58      FURNITRE         1
4          1061      Lily Sharma    M  69      Electornics         2
..          ...          ...    ...    ...          ...          ...
100         1003          Jane Iyer   male  52      Furniture         two
101         1060      Emma Fernandez  Femle  58      HomeDecor         one
102         1001      R@vi Fern@ndez  Femle  18      home decor         3
103         1014      Lisa Patel    Male  NaN      Groceries         1
104         1061      DAVID DOE      F  44         clothing         4

Price    Total  Order_Date  Payment_Mode
0      -299    -897.0    07-29-2023         NaN
1       499     499.0    2023/07/02         cash
2      -299    -598.0    2023/07/07    Credit Card
3     12000   12000.0    22/07/2023      Paytm
4     12000   24000.0    17/07/2023      Paytm
..          ...          ...    ...          ...
100      299      NaN    2023-07-22    Netbanking
101      299      NaN    18-Aug-2023    Credit Card
102      999    2997.0    2023/07/24         Cash
103     5000   5000.0    07-01-2023         NaN
104      999    3996.0    20/08/2023      debit

[105 rows x 10 columns]
```

• Standardize Gender Column

```
import pandas as pd

df = pd.read_csv("unclean_sales_data (1).csv")
print(df)
df = df.drop_duplicates()

df = df.apply(lambda x: x.str.strip() if x.dtype == "object" else x)

df["Gender"] = df["Gender"].replace({
    "M": "Male",
    "F": "Female",
    "femlae": "Female",
    "Femle": "Female",
    "MALE": "Male",
    "FEMALE": "Female",
    "female": "Female",
    "male": "Male"
})

Order_ID    Customer_Name  Gender  Age  Product_Category  Quantity  \
0          1046      Asha Iyer  FEMALE  20         clothing         3
1          1079      Lily Thom@s  femlae  45          Cloth         one
2          1054      Anne Smith  Femle  50      Electronics         2
3          1061      Lisa Smith    F  58      FURNITRE         1
4          1061      Lily Sharma    M  69      Electornics         2
..          ...          ...    ...    ...          ...          ...
100         1003          Jane Iyer   male  52      Furniture         two
101         1060      Emma Fernandez  Femle  58      HomeDecor         one
102         1001      R@vi Fern@ndez  Femle  18      home decor         3
103         1014      Lisa Patel    Male  NaN      Groceries         1
104         1061      DAVID DOE      F  44         clothing         4

Price    Total  Order_Date  Payment_Mode
0      -299    -897.0    07-29-2023         NaN
1       499     499.0    2023/07/02         cash
2      -299    -598.0    2023/07/07    Credit Card
3     12000   12000.0    22/07/2023      Paytm
4     12000   24000.0    17/07/2023      Paytm
..          ...          ...    ...          ...
100      299      NaN    2023-07-22    Netbanking
101      299      NaN    18-Aug-2023    Credit Card
102      999    2997.0    2023/07/24         Cash
103     5000   5000.0    07-01-2023         NaN
104      999    3996.0    20/08/2023      debit

[105 rows x 10 columns]
```

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• Clean Product Categories

```
import pandas as pd

df = pd.read_csv("unclean_sales_data (1).csv")
print(df)
df = df.drop_duplicates()

df = df.apply(lambda x: x.str.strip() if x.dtype == "object" else x)

df["Gender"] = df["Gender"].replace({
    "M": "Male",
    "F": "Female",
    "femlae": "Female",
    "Femle": "Female",
    "MALE": "Male",
    "FEMALE": "Female",
    "female": "Female",
    "male": "Male"
})

df["Product_Category"] = df["Product_Category"].replace({
    "clothng": "Clothing",
    "Cloth": "Clothing",
    "electronics": "Electronics",
    "Electornics": "Electronics",
    "ELECTRONICS": "Electronics",
    "FURNITRE": "Furniture",
    "Furnitre": "Furniture",
    "grocey": "Groceries",
    "GROCEY": "Groceries",
    "HOMEDECOR": "Home Decor",
    "HomeDecor": "Home Decor",
    "HOME DECOR": "Home Decor"
})
```

	Order_ID	Customer_Name	Gender	Age	Product_Category	Quantity \
0	1046	Asha Iyer	FEMALE	20	clothng	3
1	1079	Lily Thom@s	femlae	45	Cloth	one
2	1054	Anne Smith	Femle	50	Electronics	2
3	1061	Lisa Smith	F	58	FURNITRE	1
4	1061	Lily Sharma	M	69	Electornics	2
..	...	...	...	...	...	...
100	1003	Jane Iyer	male	52	Furniture	two
101	1060	Emma Fernandez	Femle	58	HomeDecor	one
102	1001	R@vi Fern@ndez	Femle	18	home decor	3
103	1014	Lisa Patel	Male	NaN	Groceries	1
104	1061	DAVID DOE	F	44	clothng	4

	Price	Total	Order_Date	Payment_Mode
0	-299	-897.0	07-29-2023	NaN
1	499	499.0	2023/07/02	cash
2	-299	-598.0	2023/07/07	Credit Card
3	12000	12000.0	22/07/2023	Paytm
4	12000	24000.0	17/07/2023	Paytm
..	...	...	...	...
100	299	NaN	2023-07-22	Netbanking
101	299	NaN	18-Aug-2023	Credit Card
102	999	2997.0	2023/07/24	Cash
103	5000	5000.0	07-01-2023	NaN
104	999	3996.0	20/08/2023	debit

[105 rows x 10 columns]

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- Convert Quantity Words into Numbers:**

```
import pandas as pd

df = pd.read_csv("unclean_sales_data (1).csv")
print(df)
df = df.drop_duplicates()

df = df.apply(lambda x: x.str.strip() if x.dtype == "object" else x)

df["Gender"] = df["Gender"].replace({
    "M": "Male",
    "F": "Female",
    "femlae": "Female",
    "Femle": "Female",
    "MALE": "Male",
    "FEMALE": "Female",
    "female": "Female",
    "male": "Male"
})

df["Product_Category"] = df["Product_Category"].replace({
    "clothing": "Clothing",
    "Cloth": "Clothing",
    "electronics": "Electronics",
    "Electornics": "Electronics",
    "ELECTRONICS": "Electronics",
    "FURNITRE": "Furniture",
    "Furnitre": "Furniture",
    "grocey": "Groceries",
    "GROCEY": "Groceries",
    "HOMEDECOR": "Home Decor",
    "HomeDecor": "Home Decor",
    "HOME DECOR": "Home Decor"
})

df["Quantity"] = df["Quantity"].replace({
    "one": 1,
    "two": 2
})
```

	Order_ID	Customer_Name	Gender	Age	Product_Category	Quantity
0	1046	Asha Iyer	FEMALE	20	clothing	3
1	1079	Lily Thom@s	femlae	45	Cloth	one
2	1054	Anne Smith	Femle	50	Electronics	2
3	1061	Lisa Smith	F	58	FURNITRE	1
4	1061	Lily Sharma	M	69	Electornics	2
..	...	...	...	...	...	...
100	1003	Jane Iyer	male	52	Furniture	two
101	1060	Emma Fernandez	Femle	58	HomeDecor	one
102	1001	R@vi Fern@ndez	Femle	18	home decor	3
103	1014	Lisa Patel	Male	NaN	Groceries	1
104	1061	DAVID DOE	F	44	clothing	4

	Price	Total	Order_Date	Payment_Mode
0	-299	-897.0	07-29-2023	NaN
1	499	499.0	2023/07/02	cash
2	-299	-598.0	2023/07/07	Credit Card
3	12000	12000.0	22/07/2023	Paytm
4	12000	24000.0	17/07/2023	Paytm
..	...	...	...	...
100	299	NaN	2023-07-22	Netbanking
101	299	NaN	18-Aug-2023	Credit Card
102	999	2997.0	2023/07/24	Cash
103	5000	5000.0	07-01-2023	NaN
104	999	3996.0	20/08/2023	debit

[105 rows x 10 columns]

DEPARTMENT OF ROBOTICS & ARTIFICIAL INTELLIGENCE

• Convert Quantity to Numeric Type:

```
import pandas as pd

df = pd.read_csv("unclean_sales_data (1).csv")
print(df)
df = df.drop_duplicates()

df = df.apply(lambda x: x.str.strip() if x.dtype == "object" else x)

df["Gender"] = df["Gender"].replace({
    "H": "Male",
    "F": "Female",
    "Femlae": "Female",
    "Femle": "Female",
    "MALE": "Male",
    "FEMALE": "Female",
    "female": "Female",
    "male": "Male"
})

df["Product_Category"] = df["Product_Category"].replace({
    "clothing": "Clothing",
    "Cloth": "Clothing",
    "electronics": "Electronics",
    "Electornics": "Electronics",
    "ELECTRONICS": "Electronics",
    "FURNITRE": "Furniture",
    "Furnitre": "Furniture",
    "grocey": "Groceries",
    "GROCEY": "Groceries",
    "HOMEDECOR": "Home Decor",
    "HomeDecor": "Home Decor",
    "HOME DECOR": "Home Decor"
})

df["Quantity"] = df["Quantity"].replace({
    "one": 1,
    "two": 2
})

df["Quantity"] = pd.to_numeric(df["Quantity"], errors="coerce")
```

	Order_ID	Customer_Name	Gender	Age	Product_Category	Quantity \
0	1046	Asha Iyer	FEMALE	20	clothing	3
1	1079	Lily Thom@s	femlae	45	Cloth	one
2	1054	Anne Smith	Femle	50	Electronics	2
3	1061	Lisa Smith	F	58	FURNITRE	1
4	1061	Lily Sharma	M	69	Electornics	2
..	...	...	...	...	...	...
100	1003	Jane Iyer	male	52	Furniture	two
101	1060	Emma Fernandez	Femle	58	HomeDecor	one
102	1001	R@vi Fern@ndez	Femle	18	home decor	3
103	1014	Lisa Patel	Male	NaN	Groceries	1
104	1061	DAVID DOE	F	44	clothing	4

	Price	Total	Order_Date	Payment_Mode
0	-299	-897.0	07-29-2023	NaN
1	499	499.0	2023/07/02	cash
2	-299	-598.0	2023/07/07	Credit Card
3	12000	12000.0	22/07/2023	Paytm
4	12000	24000.0	17/07/2023	Paytm
..	...	...	...	...
100	299	NaN	2023-07-22	Netbanking
101	299	NaN	18-Aug-2023	Credit Card
102	999	2997.0	2023/07/24	Cash
103	5000	5000.0	07-01-2023	NaN
104	999	3996.0	20/08/2023	debit

[105 rows x 10 columns]

DEPARTMENT OF ROBOTICS & ARTIFICIAL INTELLIGENCE

• Fix Age Column

```
import pandas as pd

df = pd.read_csv("unclean_sales_data (1).csv")
print(df)
df = df.drop_duplicates()

df = df.apply(lambda x: x.str.strip() if x.dtype == "object" else x)

df["Gender"] = df["Gender"].replace({
    "M": "Male",
    "F": "Female",
    "femlae": "Female",
    "Femle": "Female",
    "MALE": "Male",
    "FEHALE": "Female",
    "female": "Female",
    "male": "Male"
})

df["Product_Category"] = df["Product_Category"].replace({
    "clothing": "Clothing",
    "Cloth": "Clothing",
    "electronics": "Electronics",
    "Electornics": "Electronics",
    "ELECTRONICS": "Electronics",
    "FURNITRE": "Furniture",
    "Furnitre": "Furniture",
    "grocey": "Groceries",
    "GROCEY": "Groceries",
    "HOMEDECOR": "Home Decor",
    "HomeDecor": "Home Decor",
    "HOME DECOR": "Home Decor"
})

df["Quantity"] = df["Quantity"].replace({
    "one": 1,
    "two": 2
})

df["Quantity"] = pd.to_numeric(df["Quantity"], errors="coerce")
df["Age"] = df["Age"].replace({
    "thirty": 30,
    "forty": 40,
    "twenty": 20
})

df["Age"] = pd.to_numeric(df["Age"], errors="coerce")
```

	Order_ID	Customer_Name	Gender	Age	Product_Category	Quantity
0	1046	Asha Iyer	FEMALE	20	clothing	3
1	1079	Lily Thom@s	femlae	45	Cloth	one
2	1054	Anne Smith	Femle	50	Electronics	2
3	1061	Lisa Smith	F	58	FURNITRE	1
4	1061	Lily Sharma	M	69	Electornics	2
..	...	...	...	...	...	...
100	1003	Jane Iyer	male	52	Furniture	two
101	1060	Emma Fernandez	Femle	58	HomeDecor	one
102	1001	R@vi Fern@ndez	Femle	18	home decor	3
103	1014	Lisa Patel	Male	NaN	Groceries	1
104	1061	DAVID DOE	F	44	clothing	4

	Price	Total	Order_Date	Payment_Mode
0	-299	-897.0	07-29-2023	NaN
1	499	499.0	2023/07/02	cash
2	-299	-598.0	2023/07/07	Credit Card
3	12000	12000.0	22/07/2023	Paytm
4	12000	24000.0	17/07/2023	Paytm
..	...	...	...	...
100	299	NaN	2023-07-22	Netbanking
101	299	NaN	18-Aug-2023	Credit Card
102	999	2997.0	2023/07/24	Cash
103	5000	5000.0	07-01-2023	NaN
104	999	3996.0	20/08/2023	debit

[105 rows x 10 columns]

DEPARTMENT OF ROBOTICS & ARTIFICIAL INTELLIGENCE

- Fix Negative Prices:**

```
import pandas as pd

df = pd.read_csv("unclean_sales_data (1).csv")
print(df)
df = df.drop_duplicates()

df = df.apply(lambda x: x.str.strip() if x.dtype == "object" else x)

df["Gender"] = df["Gender"].replace({
    "M": "Male",
    "F": "Female",
    "femlae": "Female",
    "Femle": "Female",
    "MALE": "Male",
    "FEHALE": "Female",
    "female": "Female",
    "male": "Male"
})
df["Product_Category"] = df["Product_Category"].replace({
    "clothing": "Clothing",
    "Cloth": "Clothing",
    "electronics": "Electronics",
    "Electornics": "Electronics",
    "ELECTRONICS": "Electronics",
    "FURNITRE": "Furniture",
    "Furnitre": "Furniture",
    "grocey": "Groceries",
    "GROCEY": "Groceries",
    "HOMEDECOR": "Home Decor",
    "HomeDecor": "Home Decor",
    "HOME DECOR": "Home Decor"
})
df["Quantity"] = df["Quantity"].replace({
    "one": 1,
    "two": 2
})
df["Quantity"] = pd.to_numeric(df["Quantity"], errors="coerce")
df["Age"] = df["Age"].replace({
    "thirty": 30,
    "forty": 40,
    "twenty": 20
})
df["Age"] = pd.to_numeric(df["Age"], errors="coerce")
df["Price"] = df["Price"].abs()
```

	Order_ID	Customer_Name	Gender	Age	Product_Category	Quantity	\
0	1046	Asha Iyer	FEMALE	20	clothing	3	
1	1079	Lily Thom@s	femlae	45	Cloth	one	
2	1054	Anne Smith	Femle	50	Electronics	2	
3	1061	Lisa Smith	F	58	FURNITRE	1	
4	1061	Lily Sharma	M	69	Electornics	2	
..	...	...	...	...	...	...	
100	1003	Jane Iyer	male	52	Furniture	two	
101	1060	Emma Fernandez	Femle	58	HomeDecor	one	
102	1001	R@vi Fern@ndez	Femle	18	home decor	3	
103	1014	Lisa Patel	Male	NaN	Groceries	1	
104	1061	DAVID DOE	F	44	clothing	4	

	Price	Total	Order_Date	Payment_Mode
0	-299	-897.0	07-29-2023	NaN
1	499	499.0	2023/07/02	cash
2	-299	-598.0	2023/07/07	Credit Card
3	12000	12000.0	22/07/2023	Paytm
4	12000	24000.0	17/07/2023	Paytm
..	...	...	...	...
100	299	NaN	2023-07-22	Netbanking
101	299	NaN	18-Aug-2023	Credit Card
102	999	2997.0	2023/07/24	Cash
103	5000	5000.0	07-01-2023	NaN
104	999	3996.0	20/08/2023	debit

[105 rows x 10 columns]

DEPARTMENT OF ROBOTICS & ARTIFICIAL INTELLIGENCE

- Recalculate Total Column:**

```
import pandas as pd

df = pd.read_csv("unclean_sales_data (1).csv")
print(df)
df = df.drop_duplicates()

df = df.apply(lambda x: x.str.strip() if x.dtype == "object" else x)

df["Gender"] = df["Gender"].replace({
    "M": "Male",
    "F": "Female",
    "Femlae": "Female",
    "Femle": "Female",
    "MALE": "Male",
    "FEMALE": "Female",
    "female": "Female",
    "male": "Male"
})
df["Product_Category"] = df["Product_Category"].replace({
    "clothng": "Clothing",
    "Cloth": "Clothing",
    "electronics": "Electronics",
    "Electornics": "Electronics",
    "ELECTRONICS": "Electronics",
    "FURNITRE": "Furniture",
    "Furnitre": "Furniture",
    "grocey": "Groceries",
    "GROCEY": "Groceries",
    "HOMEDECOR": "Home Decor",
    "HomeDecor": "Home Decor",
    "HOME DECOR": "Home Decor"
})
df["Quantity"] = df["Quantity"].replace({
    "one": 1,
    "two": 2
})
df["Quantity"] = pd.to_numeric(df["Quantity"], errors="coerce")
df["Age"] = df["Age"].replace({
    "thirty": 30,
    "forty": 40,
    "twenty": 20
})
df["Age"] = pd.to_numeric(df["Age"], errors="coerce")
df["Price"] = df["Price"].abs()

df["Total"] = df["Quantity"] * df["Price"]
```

	Order_ID	Customer_Name	Gender	Age	Product_Category	Quantity
0	1046	Asha Iyer	FEMALE	20	clothng	3
1	1079	Lily Thom@s	femlae	45	Cloth	one
2	1054	Anne Smith	Femle	50	Electronics	2
3	1061	Lisa Smith	F	58	FURNITRE	1
4	1061	Lily Sharma	M	69	Electornics	2
..	...	...	...	...	...	...
100	1003	Jane Iyer	male	52	Furniture	two
101	1060	Emma Fernandez	Femle	58	HomeDecor	one
102	1001	R@vi Fern@ndez	Femle	18	home decor	3
103	1014	Lisa Patel	Male	NaN	Groceries	1
104	1061	DAVID DOE	F	44	clothng	4

	Price	Total	Order_Date	Payment_Mode
0	-299	-897.0	07-29-2023	NaN
1	499	499.0	2023/07/02	cash
2	-299	-598.0	2023/07/07	Credit Card
3	12000	12000.0	22/07/2023	Paytm
4	12000	24000.0	17/07/2023	Paytm
..	...	...	...	...
100	299	NaN	2023-07-22	Netbanking
101	299	NaN	18-Aug-2023	Credit Card
102	999	2997.0	2023/07/24	Cash
103	5000	5000.0	07-01-2023	NaN
104	999	3996.0	20/08/2023	debit

[105 rows x 10 columns]

DEPARTMENT OF ROBOTICS & ARTIFICIAL INTELLIGENCE

• STANDARIZE PAYMENT METHOD:

```

})
df["Product_Category"] = df["Product_Category"].replace({
    "clothng": "Clothing",
    "Cloth": "Clothing",
    "electronics": "Electronics",
    "Electornics": "Electronics",
    "ELECTRONICS": "Electronics",
    "FURNITRE": "Furniture",
    "Furnitre": "Furniture",
    "grocey": "Groceries",
    "GROCEY": "Groceries",
    "HOMEDECOR": "Home Decor",
    "HomeDecor": "Home Decor",
    "HOME DECOR": "Home Decor"
})
df["Quantity"] = df["Quantity"].replace({
    "one": 1,
    "two": 2
})
df["Quantity"] = pd.to_numeric(df["Quantity"], errors="coerce")
df["Age"] = df["Age"].replace({
    "thirty": 30,
    "forty": 40,
    "twenty": 20
})
df["Age"] = pd.to_numeric(df["Age"], errors="coerce")
df["Price"] = df["Price"].abs()

df["Total"] = df["Quantity"] * df["Price"]

df["Payment_Mode"] = df["Payment_Mode"].replace({
    "cash": "Cash",
    "upi": "UPI",
    "debit": "Debit Card",
    "Credit Card": "Credit Card"
})

```

	Order_ID	Customer_Name	Gender	Age	Product_Category	Quantity
0	1046	Asha Iyer	FEMALE	20	clothng	3
1	1079	Lily Thom@s	femlae	45	Cloth	one
2	1054	Anne Smith	Femle	50	Electronics	2
3	1061	Lisa Smith	F	58	FURNITRE	1
4	1061	Lily Sharma	M	69	Electornics	2
..	...	...	...	...	...	...
100	1003	Jane Iyer	male	52	Furniture	two
101	1060	Emma Fernandez	Femle	58	HomeDecor	one
102	1001	R@vi Fern@ndez	Femle	18	home decor	3
103	1014	Lisa Patel	Male	NaN	Groceries	1
104	1061	DAVID DOE	F	44	clothng	4

	Price	Total	Order_Date	Payment_Mode
0	-299	-897.0	07-29-2023	NaN
1	499	499.0	2023/07/02	cash
2	-299	-598.0	2023/07/07	Credit Card
3	12000	12000.0	22/07/2023	Paytm
4	12000	24000.0	17/07/2023	Paytm
..	...	...	...	...
100	299	NaN	2023-07-22	Netbanking
101	299	NaN	18-Aug-2023	Credit Card
102	999	2997.0	2023/07/24	Cash
103	5000	5000.0	07-01-2023	NaN
104	999	3996.0	20/08/2023	debit

[105 rows x 10 columns]

DEPARTMENT OF ROBOTICS & ARTIFICIAL INTELLIGENCE

• **COVERTING DATES:**

```

    one : 1,
    "two": 2
})
df["Quantity"] = pd.to_numeric(df["Quantity"], errors="coerce")
df["Age"] = df["Age"].replace({
    "thirty": 30,
    "forty": 40,
    "twenty": 20
})
df["Age"] = pd.to_numeric(df["Age"], errors="coerce")
df["Price"] = df["Price"].abs()

df["Total"] = df["Quantity"] * df["Price"]

df["Payment_Mode"] = df["Payment_Mode"].replace({
    "cash": "Cash",
    "upi": "UPI",
    "debit": "Debit Card",
    "Credit Card": "Credit Card"
})
df["Order_Date"] = pd.to_datetime(df["Order_Date"], errors="coerce")

```

	Order_ID	Customer_Name	Gender	Age	Product_Category	Quantity	\
0	1046	Asha Iyer	FEMALE	20	clothing	3	
1	1079	Lily Thom@s	femlae	45	Cloth	one	
2	1054	Anne Smith	Femle	50	Electronics	2	
3	1061	Lisa Smith	F	58	FURNITRE	1	
4	1061	Lily Sharma	M	69	Electornics	2	
..	...	...	...	...	...	...	
100	1003	Jane Iyer	male	52	Furniture	two	
101	1060	Emma Fernandez	Femle	58	HomeDecor	one	
102	1001	R@vi Fern@ndez	Femle	18	home decor	3	
103	1014	Lisa Patel	Male	NaN	Groceries	1	
104	1061	DAVID DOE	F	44	clothing	4	

	Price	Total	Order_Date	Payment_Mode
0	-299	-897.0	07-29-2023	NaN
1	499	499.0	2023/07/02	cash
2	-299	-598.0	2023/07/07	Credit Card
3	12000	12000.0	22/07/2023	Paytm
4	12000	24000.0	17/07/2023	Paytm
..	...	...	...	...
100	299	NaN	2023-07-22	Netbanking
101	299	NaN	18-Aug-2023	Credit Card
102	999	2997.0	2023/07/24	Cash
103	5000	5000.0	07-01-2023	NaN
104	999	3996.0	20/08/2023	debit

[105 rows x 10 columns]

DEPARTMENT OF ROBOTICS & ARTIFICIAL INTELLIGENCE

• **Remove Special Characters from Names**

```

    "Twenty": 20
})
df["Age"] = pd.to_numeric(df["Age"], errors="coerce")
df["Price"] = df["Price"].abs()

df["Total"] = df["Quantity"] * df["Price"]

df["Payment_Mode"] = df["Payment_Mode"].replace({
    "cash": "Cash",
    "upi": "UPI",
    "debit": "Debit Card",
    "Credit Card": "Credit Card"
})
df["Order_Date"] = pd.to_datetime(df["Order_Date"], errors="coerce")
df["Customer_Name"] = df["Customer_Name"].str.replace(r'[^a-zA-Z ]', '', regex=True)

```

	Order_ID	Customer_Name	Gender	Age	Product_Category	Quantity \
0	1046	Asha Iyer	FEMALE	20	clothing	3
1	1079	Lily Thom@s	femlae	45	Cloth	one
2	1054	Anne Smith	Femle	50	Electronics	2
3	1061	Lisa Smith	F	58	FURNITRE	1
4	1061	Lily Sharma	M	69	Electornics	2
..	...	...	...	...	...	...
100	1003	Jane Iyer	male	52	Furniture	two
101	1060	Emma Fernandez	Femle	58	HomeDecor	one
102	1001	R@vi Fern@ndez	Femle	18	home decor	3
103	1014	Lisa Patel	Male	NaN	Groceries	1
104	1061	DAVID DOE	F	44	clothing	4

	Price	Total	Order_Date	Payment_Mode
0	-299	-897.0	07-29-2023	NaN
1	499	499.0	2023/07/02	cash
2	-299	-598.0	2023/07/07	Credit Card
3	12000	12000.0	22/07/2023	Paytm
4	12000	24000.0	17/07/2023	Paytm
..	...	...	...	...
100	299	NaN	2023-07-22	Netbanking
101	299	NaN	18-Aug-2023	Credit Card
102	999	2997.0	2023/07/24	Cash
103	5000	5000.0	07-01-2023	NaN
104	999	3996.0	20/08/2023	debit

[105 rows x 10 columns]

DEPARTMENT OF ROBOTICS & ARTIFICIAL INTELLIGENCE

• **Fill Missing Values**

```
df["Age"] = df["Age"].replace({
    "thirty": 30,
    "forty": 40,
    "twenty": 20
})
df["Age"] = pd.to_numeric(df["Age"], errors="coerce")
df["Price"] = df["Price"].abs()

df["Total"] = df["Quantity"] * df["Price"]

df["Payment_Mode"] = df["Payment_Mode"].replace({
    "cash": "Cash",
    "upi": "UPI",
    "debit": "Debit Card",
    "Credit Card": "Credit Card"
})
df["Order_Date"] = pd.to_datetime(df["Order_Date"], errors="coerce")
df["Customer_Name"] = df["Customer_Name"].str.replace(r'^a-zA-Z ', '', regex=True)
df["Gender"] = df["Gender"].fillna("Unknown")
df["Payment_Mode"] = df["Payment_Mode"].fillna("Unknown")
```

	Order_ID	Customer_Name	Gender	Age	Product_Category	Quantity	\
0	1046	Asha Iyer	FEMALE	20	clothing	3	
1	1079	Lily Thom@s	femlae	45	Cloth	one	
2	1054	Anne Smith	Femle	50	Electronics	2	
3	1061	Lisa Smith	F	58	FURNITRE	1	
4	1061	Lily Sharma	M	69	Electornics	2	
..	...	...	...	...	...	...	
100	1003	Jane Iyer	male	52	Furniture	two	
101	1060	Emma Fernandez	Femle	58	HomeDecor	one	
102	1001	R@vi Fern@ndez	Femle	18	home decor	3	
103	1014	Lisa Patel	Male	NaN	Groceries	1	
104	1061	DAVID DOE	F	44	clothing	4	

	Price	Total	Order_Date	Payment_Mode
0	-299	-897.0	07-29-2023	NaN
1	499	499.0	2023/07/02	cash
2	-299	-598.0	2023/07/07	Credit Card
3	12000	12000.0	22/07/2023	Paytm
4	12000	24000.0	17/07/2023	Paytm
..	...	...	...	...
100	299	NaN	2023-07-22	Netbanking
101	299	NaN	18-Aug-2023	Credit Card
102	999	2997.0	2023/07/24	Cash
103	5000	5000.0	07-01-2023	NaN
104	999	3996.0	20/08/2023	debit

[105 rows x 10 columns]

DEPARTMENT OF ROBOTICS & ARTIFICIAL INTELLIGENCE

• **Remove Rows with Important Missing Data**

```
df["Age"] = pd.to_numeric(df["Age"], errors="coerce")
df["Price"] = df["Price"].abs()

df["Total"] = df["Quantity"] * df["Price"]

df["Payment_Mode"] = df["Payment_Mode"].replace({
    "cash": "Cash",
    "upi": "UPI",
    "debit": "Debit Card",
    "Credit Card": "Credit Card"
})

df["Order_Date"] = pd.to_datetime(df["Order_Date"], errors="coerce")
df["Customer_Name"] = df["Customer_Name"].str.replace(r'^[a-zA-Z ]', '', regex=True)
df["Gender"] = df["Gender"].fillna("Unknown")
df["Payment_Mode"] = df["Payment_Mode"].fillna("Unknown")
df = df.dropna(subset=["Customer_Name", "Age", "Quantity"])
```

	Order_ID	Customer_Name	Gender	Age	Product_Category	Quantity \
0	1046	Asha Iyer	FEMALE	20	clothng	3
1	1079	Lily Thom@s	femlae	45	Cloth	one
2	1054	Anne Smith	Femle	50	Electronics	2
3	1061	Lisa Smith	F	58	FURNITRE	1
4	1061	Lily Sharma	M	69	Electornics	2
..	...	...	...	...	...	...
100	1003	Jane Iyer	male	52	Furniture	two
101	1060	Emma Fernandez	Femle	58	HomeDecor	one
102	1001	R@vi Fern@ndez	Femle	18	home decor	3
103	1014	Lisa Patel	Male	NaN	Groceries	1
104	1061	DAVID DOE	F	44	clothng	4

	Price	Total	Order_Date	Payment_Mode
0	-299	-897.0	07-29-2023	NaN
1	499	499.0	2023/07/02	cash
2	-299	-598.0	2023/07/07	Credit Card
3	12000	12000.0	22/07/2023	Paytm
4	12000	24000.0	17/07/2023	Paytm
..	...	...	...	...
100	299	NaN	2023-07-22	Netbanking
101	299	NaN	18-Aug-2023	Credit Card
102	999	2997.0	2023/07/24	Cash
103	5000	5000.0	07-01-2023	NaN
104	999	3996.0	20/08/2023	debit

[105 rows x 10 columns]

DEPARTMENT OF ROBOTICS & ARTIFICIAL INTELLIGENCE

• **DISPLAY CLEANED DATA SET:**

	Order_ID	Customer_Name	Gender	Age	Product_Category	Quantity	\
0	1046	Asha Iyer	FEMALE	20	clothing	3	
1	1079	Lily Thom@s	femlae	45	Cloth	one	
2	1054	Anne Smith	Femle	50	Electronics	2	
3	1061	Lisa Smith	F	58	FURNITRE	1	
4	1061	Lily Sharma	M	69	Electornics	2	
..	...	...	...	...	...	...	
100	1003	Jane Iyer	male	52	Furniture	two	
101	1060	Emma Fernandez	Femle	58	HomeDecor	one	
102	1001	R@vi Fern@ndez	Femle	18	home decor	3	
103	1014	Lisa Patel	Male	NaN	Groceries	1	
104	1061	DAVID DOE	F	44	clothing	4	

	Price	Total	Order_Date	Payment_Mode
0	-299	-897.0	07-29-2023	NaN
1	499	499.0	2023/07/02	cash
2	-299	-598.0	2023/07/07	Credit Card
3	12000	12000.0	22/07/2023	Paytm
4	12000	24000.0	17/07/2023	Paytm
..	...	...	...	...
100	299	NaN	2023-07-22	Netbanking
101	299	NaN	18-Aug-2023	Credit Card
102	999	2997.0	2023/07/24	Cash
103	5000	5000.0	07-01-2023	NaN
104	999	3996.0	20/08/2023	debit

[105 rows x 10 columns]

	Order_ID	Customer_Name	Gender	Age	Product_Category	Quantity	\
0	1046	Asha Iyer	Female	20.0	Clothing	3.0	
1	1079	Lily Thoms	Female	45.0	Clothing	1.0	
2	1054	Anne Smith	Female	50.0	Electronics	2.0	
3	1061	Lisa Smith	Female	58.0	Furniture	1.0	
4	1061	Lily Sharma	Male	69.0	Electronics	2.0	
..	...	...	...	...	...	...	
97	1007	jane kumar	Female	23.0	Electronics	2.0	
98	1017	Lily Verma	Unknown	29.0	Clothing	4.0	
100	1003	Jane Iyer	Male	52.0	Furniture	2.0	
101	1060	Emma Fernandez	Female	58.0	Home Decor	1.0	
102	1001	Rvi Fernndez	Female	18.0	home decor	3.0	

	Price	Total	Order_Date	Payment_Mode
0	299	897.0	2023-07-29	Unknown
1	499	499.0	NaN	Cash
2	299	598.0	NaN	Credit Card
3	12000	12000.0	NaN	Paytm
4	12000	24000.0	NaN	Paytm
..	...	...	...	...
97	499	998.0	NaN	Paytm
98	5000	20000.0	NaN	Paytm
100	299	598.0	NaN	Netbanking
101	299	299.0	NaN	Credit Card
102	999	2997.0	NaN	Cash

[91 rows x 10 columns]

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- USING MATLIB

```
import pandas as pd
import matplotlib.pyplot as plt
```

```
df = pd.read_csv("unclean_sales_data (1).csv")
print(df)
df = df.drop_duplicates()

df = df.apply(lambda x: x.str.strip() if x.dtype == "object" else x)

df["Gender"] = df["Gender"].replace({
    "M": "Male",
    "F": "Female",
    "femlae": "Female",
    "Femle": "Female",
    "MALE": "Male",
    "FEMALE": "Female",
    "female": "Female",
    "male": "Male"
})

df["Product_Category"] = df["Product_Category"].replace({
    "clothng": "Clothing",
    "Cloth": "Clothing",
    "electronics": "Electronics",
    "Electornics": "Electronics",
    "ELECTRONICS": "Electronics",
    "FURNITRE": "Furniture",
    "Furnitre": "Furniture",
    "grocey": "Groceries",
    "GROCEY": "Groceries",
    "HOMEDECOR": "Home Decor",
    "HomeDecor": "Home Decor",
    "HOME DECOR": "Home Decor"
})

df["Quantity"] = df["Quantity"].replace({
    "one": 1,
    "two": 2
})

df["Quantity"] = pd.to_numeric(df["Quantity"], errors="coerce")
df["Age"] = df["Age"].replace({
    "thirty": 30,
    "forty": 40,
    "twenty": 20
})

df["Age"] = pd.to_numeric(df["Age"], errors="coerce")
df["Price"] = df["Price"].abs()
```

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```
df["Total"] = df["Quantity"] * df["Price"]

df["Payment_Mode"] = df["Payment_Mode"].replace({
    "cash": "Cash",
    "upi": "UPI",
    "debit": "Debit Card",
    "Credit Card": "Credit Card"
})

df["Order_Date"] = pd.to_datetime(df["Order_Date"], errors="coerce")
df["Customer_Name"] = df["Customer_Name"].str.replace(r'^a-zA-Z ', '', regex=True)
df["Gender"] = df["Gender"].fillna("Unknown")
df["Payment_Mode"] = df["Payment_Mode"].fillna("Unknown")
df = df.dropna(subset=["Customer_Name", "Age", "Quantity"])
print(df)

df["Product_Category"].value_counts().plot(kind="bar")

plt.title("Product Category Count")
plt.xlabel("Category")
plt.ylabel("Count")

plt.show()
```

	Order_ID	Customer_Name	Gender	Age	Product_Category	Quantity
0	1046	Asha Iyer	FEMALE	20	clothing	3
1	1079	Lily Thom@s	femlae	45	Cloth	one
2	1054	Anne Smith	Femle	50	Electronics	2
3	1061	Lisa Smith	F	58	FURNITRE	1
4	1061	Lily Sharma	M	69	Electornics	2
..	...	...	...	...	...	...
100	1003	Jane Iyer	male	52	Furniture	two
101	1060	Emma Fernandez	Femle	58	HomeDecor	one
102	1001	R@vi Fern@ndez	Femle	18	home decor	3
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98	5000	20000.0	NaN	Paytm
100	299	598.0	NaN	Netbanking
101	299	299.0	NaN	Credit Card
102	999	2997.0	NaN	Cash

[91 rows x 10 columns]

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